

Data Analytics Using Python

Instructions for Mini Project

Objective

To apply concepts of **data preprocessing**, **exploratory data analysis (EDA)**, **statistical inference**, and **predictive modeling** on a real-world dataset using Python.

Step 1: Problem Definition & Dataset Selection

1. Choose any **real-world dataset** (from sources like Kaggle, UCI ML Repository, data.gov, GitHub Datasets, etc.).
2. Clearly **define your objective** — for example:
 - Predicting house prices
 - Analyzing customer churn
 - Exploring factors influencing student performance
3. **Mention the dataset source** in your Jupyter Notebook (include link and short description).
4. Describe:
 - Dataset overview (rows, columns, features)
 - Type of variables (numerical, categorical, date, etc.)

Step 2: Data Cleaning & Preparation

Perform the following preprocessing steps (if applicable):

1. **Check for missing values** and handle them appropriately (imputation or removal).
2. **Remove duplicates** and irrelevant columns.
3. **Handle outliers** using suitable methods (IQR, z-score, etc.).

4. **Encode categorical data** (Label Encoding / One-Hot Encoding).
5. **Normalize or scale** data if required.
6. Provide **brief justification** for each preprocessing step.

Step 3: Exploratory Data Analysis (EDA)

Perform EDA to **understand data patterns and relationships** (whatever applicable):

1. **Univariate analysis** — histograms, boxplots, distribution plots.
2. **Bivariate/multivariate analysis** — pairplots, correlation heatmaps.
3. **Use descriptive statistics** (mean, median, std, skewness, kurtosis).
4. Identify **patterns or insights** from the data (e.g., correlations, trends, anomalies).
5. Comment on **data distribution** and discuss why it matters for analysis.

Step 4: Statistical Analysis & Hypothesis Testing

Use inferential statistics to support insights:

1. Formulate **a hypothesis** (null & alternative).
2. Choose an appropriate **statistical test** (t-test, ANOVA, chi-square, etc.).
3. Compute and interpret:
 - **p-value** → for significance
 - **Confidence interval** → for reliability of estimates
4. Discuss **Type I and Type II errors** and their relevance in your context.

Step 5: Modeling and Pattern Discovery

Perform one of the following analytical techniques:

Option A: Regression

- Apply **Linear Regression** or **logistic regression**
- Evaluate models using metrics such as R^2 , RMSE, and MAE.

Option B: Clustering or classification

- Use **K-Means** clustering or **KNN**
- Visualize clusters and interpret their significance.
- Explain how clustering helps in pattern recognition.

Step 6: Interpretation & Inference

1. Summarize major findings from EDA and modeling.
2. Discuss the **insights, implications, and business/research significance**.
3. Reflect on:
 - What patterns or relationships did you identify?
 - How did statistical testing validate your observations?
 - What could be improved with more data or features?

Step 7: Visualization and Presentation

1. Use **Matplotlib**, or **Seaborn**, to create meaningful visualizations.
2. Ensure every figure has a title, **axis labels**, and **short caption**.

Step 8: Conclusion

1. Provide a clear **summary of findings and conclusions**.
2. Discuss **limitations** and possible **future improvements**.

3. End with **one paragraph of key takeaways** — connecting data patterns, statistical tests, and model results.

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Deliverables

- A **Jupyter Notebook** containing:
 - Step-wise code with explanations and comments.
 - Visualizations and observations.
 - Properly formatted headings and markdowns.
- A short **report (1-2 pages)** summarizing:
 - Objective, data source, methods used, and key insights.